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License No.: 4864

Protocol generation date:
No.:

LABORATORY TEST PROTOCOL

Patient code:
Age:
Sex:

Sample type: dried blood spot (DBS)
Sample collected:
Sample received:
Analysis date:

Index/ Ratio	Result, Measurement Unit	Interpretation
Omega-3 index	%	
Omega-6: omega-3 ratio		
AA:EPA ratio		
Trans fat index	%	
Omega-9 index	%	

Analysis method: gas chromatography (GC)
Employee license No.

This test is not intended to diagnose, treat, cure, or prevent any disease or replace the medical advice and/or treatment obtained from a qualified healthcare practitioner.

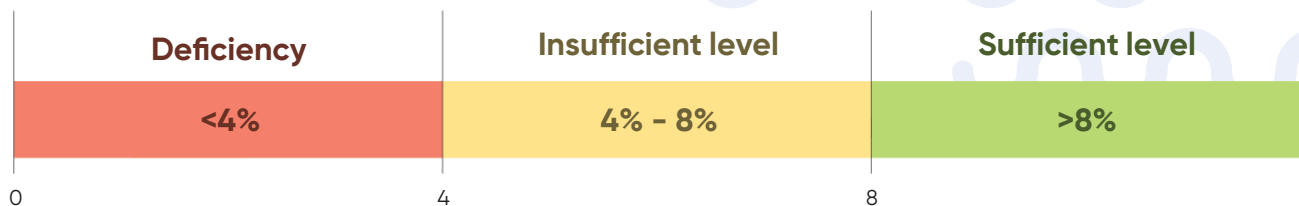
Welcome to the world of fatty acids!

We appreciate your commitment to better understanding your health and are here to help you interpret your results and make even complex medical information clear and easy to understand.

This test provides a detailed analysis of your blood fatty acid profile, reflecting your body's fatty acid balance over the past four months—similar to how the HbA1c test reflects long-term blood glucose levels.

By understanding your key fatty acid markers, you can make targeted adjustments to your diet or supplementation, if needed, to help achieve fatty acid levels that support optimal health. *Consult a qualified healthcare professional before making any changes to your diet or supplement regimen.*

Your Omega-3 (ω -3) index RESULT



The Omega-3 Index measures whether your body is receiving adequate amounts of two essential omega-3 fatty acids: eicosapentaenoic acid (EPA) and docosahexaenoic acid (DHA) [1]. It reflects the amount of EPA and DHA incorporated into the membranes of red blood cells over the past several months, providing a reliable measure of your long-term omega-3 status. A higher Omega-3 Index is associated with better cardiovascular health, supports normal brain function and overall well-being, and contributes to maintaining healthy vision and joint function [2–6].

What does your result mean?

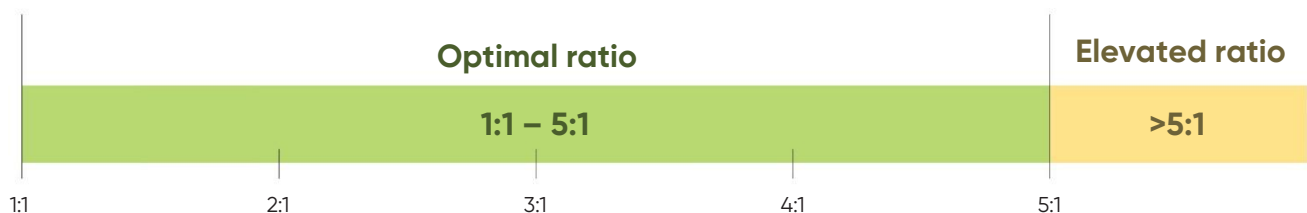
- **8–12%** – Considered optimal and associated with better overall health.
- **4–8%** – Considered a moderate level.
- **Less than 4%** – May indicate an insufficient omega-3 status [7].

How can you improve your Omega-3 Index?

- Eat more omega-3-rich foods, particularly oily fish and seafood.
- Reduce your intake of highly processed foods.
- If needed, consider omega-3 supplements, such as fish oil, algae-derived omega-3 supplements, or omega-3-fortified functional foods.

Your omega-6:omega-3 (ω -6: ω -3) ratio RESULT

The sum of all the omega-6 fatty acids is divided by the sum of all the omega-3 fatty acids.



Like omega-3 fatty acids, omega-6 fatty acids are essential and must be obtained through the diet. They are naturally found in certain vegetable oils, nuts, and seeds [8]. Omega-6 fatty acids are important for maintaining cell structure and supporting normal body function, but maintaining a healthy balance between omega-6 and omega-3 fatty acids is equally important.

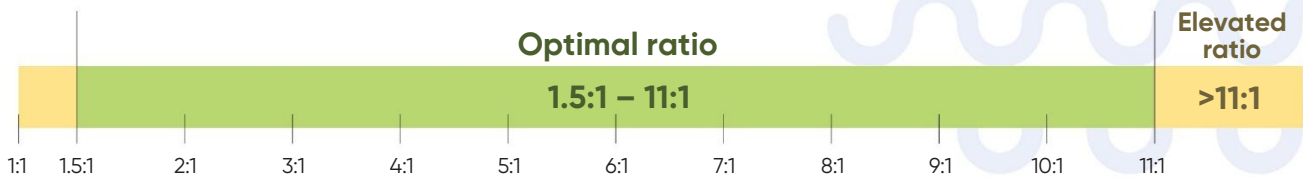
What does your result mean?

- Less than 5:1: Considered a healthy omega-6 to omega-3 (ω -6: ω -3) ratio and associated with better overall health [9].
- A higher ratio: Associated with increased inflammatory processes in the body [9, 10].

How can you improve your omega-6 to omega-3 ratio?

- Increase your intake of omega-3-rich foods.
- Reduce your intake of highly processed vegetable oils and ultra-processed foods.

AA:EPA ratio RESULT



This ratio reflects the balance between the omega-6 fatty acid arachidonic acid (AA) and the omega-3 fatty acid eicosapentaenoic acid (EPA) in your body. Arachidonic acid (AA) is an omega-6 fatty acid associated with inflammatory processes, while eicosapentaenoic acid (EPA) is an omega-3 fatty acid that helps regulate and resolve excessive inflammation. Maintaining an appropriate balance between these fatty acids is therefore important for supporting overall health.

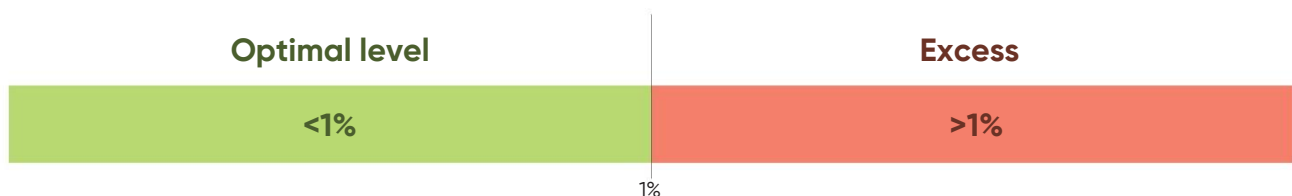
What does your result mean?

- A ratio between 1.5:1 and 11:1 is generally considered more favorable.
- A higher ratio is associated with increased inflammatory activity and a greater risk of coronary heart disease, myocardial infarction, stroke, heart failure, and peripheral vascular disease [11,12,13].

How can you improve your AA:EPA ratio?

- Increase your intake of foods rich in omega-3 fatty acids, particularly EPA and DHA, by eating oily fish such as salmon, sardines, mackerel, or herring at least 2–3 times per week;
- Consider taking omega-3 supplements if needed;
- Reduce your intake of ultra-processed foods and trans fats;
- Limit excessive intake of omega-6 fatty acids, particularly from processed vegetable oils.

Trans fat index RESULT



What does your result mean?

The Trans Fat Index reflects the proportion of trans fatty acids 18:1 and 18:2 among all fatty acids in red blood cell membranes over the past several months. A trans fat level below 1% is considered favorable for health.

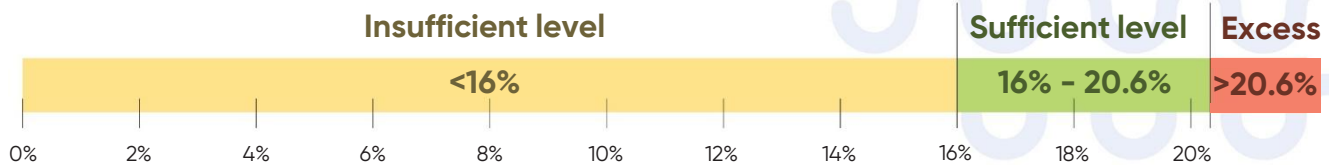
Trans fats are primarily formed during the industrial processing of fats and are commonly found in ultra-processed foods, baked goods, chips, sweets, and fast food.

Studies show that industrial trans fats may promote inflammation, cellular stress, and fatty liver disease [14,15]. These effects may contribute to the development of atherosclerosis and metabolic disorders such as insulin resistance [16].

How can you reduce your Trans fat level?

- Limit your intake of ultra-processed foods;
- Choose fresh, minimally processed foods and prepare meals at home more often.

Omega-9 index (ω -9) RESULT



What does your result mean?

The Omega-9 Index reflects the combined proportion of three monounsaturated omega-9 fatty acids—oleic acid, eicosenoic acid, and nervonic acid—among all fatty acids. A desirable omega-9 fatty acid level in red blood cell membranes is generally between 16.0% and 20.6% of total fatty acids, although slight variation may occur.

Omega-9 fatty acids are among the main monounsaturated fatty acids found in both plant- and animal-based foods. Unlike omega-3 and omega-6 fatty acids, they can be synthesized by the body, although not always in sufficient amounts. Some omega-9 fatty acids, such as nervonic acid, have anti-inflammatory properties and help support immune function and cytokine balance, while oleic acid may have beneficial effects on mood and help maintain energy levels [17,18,19].

How can you maintain an optimal Omega-9 level?

- Choose whole, minimally processed foods;
- Include adequate amounts of extra virgin olive oil, nuts, seeds, and avocados in your diet.

YOUR UNIQUE FATTY ACID PROFILE

This table presents the percentage levels of all 26 fatty acids measured in this test. It is one of the most detailed laboratory fatty acid analyses available, providing valuable insight into your fatty acid profile and the balance of fatty acids in your body.

Saturated fatty acids			
No.	Fatty acids	Your personal level, %	Reference range, %
1	Lauric acid		0.16–2.03
2	Myristic acid		0.1–2.45
3	Pentadecylic acid		0.14–0.3
4	Palmitic acid		19–27
5	Margaric acid		0.22–0.37
6	Stearic acid		8.4–15
7	Arachidic acid		0.22–0.35
8	Behenic acid		0.2–1.59
9	Tricosylic acid		0–0.78
10	Lignoceric acid		0.2–1.92
Unsaturated fatty acids			
No.	Omega-3	Your personal level, %	Reference range, %
11	α -Linolenic acid (ALA)		0.09–1
12	Eicosapentaenoic acid (EPA)		0.2–2.3
13	Docosapentaenoic acid (DPA)		0.8–1.8
14	Docosahexaenoic acid (DHA)		1.2–3.9
Omega-6			
15	Linoleic acid		18.6–29.5
16	Gamma linolenic acid (GLA)		0.03–0.13
17	Eicosadienoic (EDA)		0–0.26
18	Dihomo-gamma-linolenic acid (DGLA)		1.2–5
19	Arachidonic acid (AA)		5.2–12.9
Omega-9			
20	Oleic acid		14.2–29.5
21	Gondoic acid		0.1–0.77
22	Nervonic acid		0.13–1.96
Omega-7			
23	Palmitoleic acid		0.1–2.45
Trans fatty acids			
24	Linolelaidic acid		0.07–0.92
25	Trans Oleic acid		0–0.59
26	Trans Palmitoleic acid		0.1–2.45

What diet helps maintain an optimal fatty acid balance?



Fatty acids to limit or avoid

- Trans fats found in ultra-processed foods. Their intake should be kept as low as possible or avoided altogether.
- Excessive intake of saturated fatty acids, as it may increase low-density lipoprotein (LDL) cholesterol levels and the risk of cardiovascular disease [20,21].

What are trans fats?

Trans fats are formed during the industrial hydrogenation of oils. Partially hydrogenated vegetable oils are commonly used in the production of margarine, baked goods, sweets, and fast food. These fats help extend the shelf life of food products and allow them to be reheated repeatedly. However, they are harmful to health and have been associated with a significantly increased risk of coronary heart disease and related mortality [22].



It is recommended to choose nutrient-rich, minimally processed foods that are rich in monounsaturated (MUFAs) and polyunsaturated fatty acids (PUFAs) [23,24].

Fish and seafood are particularly rich sources of the polyunsaturated omega-3 fatty acids EPA and DHA.

Fish and Seafood (85 g serving)	EPA + DHA (mg)
Pacific and Atlantic herring	1708-1801
Atlantic salmon (wild)	1569
Atlantic salmon (farmed)	1826
Coho Salmon (farmed)	1087
Coho Salmon (wild)	900
Bluefin Tuna	1278
Skipjack Tuna	278
Pink Salmon	524
Rainbow Trout (farmed)	744
Atlantic Sardines (canned in oil)	835
Swordfish	764
Sea Bass	648
Atlantic Pollock	460
Flatfish (flounder and sole species)	255
Atlantic and Pacific Halibut	200
Oysters (farmed)	374
Alaska King Crab	351
King Mackerel	341
Walleye	338
Scallops	149
Mixed Shrimp	235
Catfish (wild)	201
Catfish (farmed)	76
Atlantic and Pacific Cod	134 - 136

Values are based on USDA FoodData Central. Unless otherwise specified, values refer to foods cooked using dry heat. Shrimp and king crab values are based on moist-heat preparation. For more detailed nutrient information, please refer to USDA FoodData Central [25].

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